PRE-ALGEBRA (8TH) · UNIT 4: SYSTEMS OF LINEAR EQUATIONS

Food Truck Face-Off

Use systems of equations to compare food trucks and find break-even points.

CP1

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Two food trucks want a spot at the spring festival: a taco truck and a burger truck. Each truck has a daily rental fee and a cost for each item it makes, and the festival committee wants to know which truck will do better.

You will write and solve systems of equations to compare costs, solve customer orders, and recommend a plan.



Taco Truck

Rent $120 per day

Cost $1 per taco

Price $3 per taco

Burger Truck

Rent $80 per day

Cost $2 per burger

Price $5 per burger

Learning Goals

- I can write and solve systems by graphing, substitution, and elimination.
- I can recognize systems with no solution or infinitely many solutions.
- I can use systems to compare profits and make a recommendation.

Project Timeline

When	What you will do
Day 1	Costs & Break-Even
Day 2	Orders & Elimination
Day 3	Profit Showdown
Final	Food Truck Business Report + reflection

Part 1: Costs & Break-Even

Day 1 · Use the truck cards. Let x be the number of items.

1. Solve the cost system y = x + 120 and y = 2x + 80 by substitution, and interpret the solution.

2. Find each truck's break-even point (revenue = cost).

3. Graph both cost lines. For how many items is the taco truck's cost lower?

Part 2: Orders & Elimination

Day 2 · Solve each system.

4. A group buys 14 items for $54 (tacos $3, burgers $5). How many of each?

5. Solve by elimination: $4t + 3b = 39$ and $2t + 5b = 37$.

6. How many solutions does each system have? (a) $y = 3x + 2$ and $6x - 2y = -4$ (b) $y = 2x + 1$ and $y = 2x - 5$

Part 3: Profit Showdown

Day 3 · Profit = price − cost per item, times the number sold, minus rent.

7. Write each truck's profit equation and decide whether the profits are ever equal for the same number of items.

8. The taco truck sells 3 tacos for every burger the burger truck sells. If the burger truck sells b burgers, for which b does the taco truck earn more?

9. Write a recommendation to the committee using your results.

Final Product: Food Truck Business Report

Your final product must include:

- Cost and break-even analysis
- Solved orders by substitution and elimination
- Examples of no-solution and infinite-solution systems
- Profit comparison with an inequality
- Business report with a recommendation

Reflection

R1. What do parallel lines mean in a business setting?

R2. When is elimination faster than substitution?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Systems are set up and solved correctly with all methods, and solution types are identified. Goes beyond the minimum with a creative, realistic choice.	Systems are set up and solved correctly with all methods, and solution types are identified.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Business report explains costs, break-even points, and the recommendation with supporting work. Polished, organized, and visually engaging.	Business report explains costs, break-even points, and the recommendation with supporting work. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.